Team Contribution Log
& Work Distribution

Performance Evaluation of Vector Indexing Techniques for AI Search Systems

Master of Applied Computing • University of Windsor

Team Member	Role / Focus Area
Devu Babu Sheeja	Data & Embeddings Lead
Ashwin Senthur Pandian	Vector Indexing Lead
Rishi Gaurangkumar Patel	Query & Evaluation Lead
Nikhil Goud Nathi	Experiments, Tuning & Visualization Lead

How this log works

Work is split into four clearly separated lanes so no two members duplicate the same task. Each member owns a distinct set of code modules and also contributes technical design and written documentation, so coding and technical work are shared evenly across the team. Shared deliverables (proposal, report, presentation, testing) are done collaboratively.

Tracking: each member maintains their own weekly log below from Week 1, recording actual dates and status. The same breakdown is mirrored on our project board (GitHub Projects) so progress is visible to the team and instructor. Entries should be kept accurate, as this log supports individual performance evaluation.

Work distribution (no overlapping tasks)

Devu Babu Sheeja — Data & Embeddings Lead

Code modules owned	data_ingestion.py, prepare_headlines.py, embeddings.py
Coding	Text loading, cleaning, normalization and segmentation; dataset preparation script; Sentence-BERT and offline embedding backends.
Technical work	Dataset selection and justification (ABC News Headlines); preprocessing pipeline; deterministic / reproducible embeddings.
Writing / docs	Dataset & preprocessing sections of the report; data methodology; Introduction / Problem Statement (proposal).

Ashwin Senthur Pandian — Vector Indexing Lead

Code modules owned	indexing.py (Flat, IVF, HNSW)
Coding	FAISS index wrappers: exact Flat baseline, IVF (k-means partitioning, nprobe), HNSW (graph M / efConstruction / efSearch); memory and build-time measurement.
Technical work	FAISS configuration; index parameter defaults; nlist auto-scaling; structural analysis of IVF clusters and HNSW graph.
Writing / docs	Indexing methodology section; Related Work on ANN methods; Novelty & Contributions (proposal).

Rishi Gaurangkumar Patel — Query & Evaluation Lead

Code modules owned	query_engine.py, benchmark.py
Coding	Query execution with per-query timing (mean / p50 / p95) and warm-up; recall@K against the Flat ground truth; benchmark orchestration (single + scalability); CSV output.
Technical work	Metric design; ground-truth pipeline; reproducibility / consistency investigation of results.
Writing / docs	Evaluation Metrics section; results analysis; Research Objectives (proposal).

Nikhil Goud Nathi — Experiments, Tuning & Visualization Lead

Code modules owned	run_benchmark.py, visualization.py, tune_parameters.py, configs/
Coding	CLI + YAML config system; comparative and scalability charts; parameter-tuning sweep (nprobe / efSearch); experiment configs.
Technical work	Scalability experiment design; parameter-tuning experiments; presentation-grade figures; end-to-end integration.
Writing / docs	System Design & Implementation Plan (proposal); README; assembly of report and presentation deck.

Shared by all members

- Proposal writing, literature review, and weekly instructor meetings.
- Integration testing, code review, and reproducibility checks.
- Final report, presentation slides, and rehearsal.

Weekly contribution logs (from Week 1)

Phases follow the proposal timeline: P1 Data & Preprocessing, P2 Embedding, P3 Index Construction, P4 Benchmarking & Evaluation, P5 Analysis & Documentation. Fill in the Date(s) and Status columns as you complete each item.

Devu Babu Sheeja — Data & Embeddings Lead

Wk	Date(s)	Tasks & contributions	Status / notes
1		Project kickoff; co-wrote proposal (Intro, Problem Statement); researched candidate datasets; set up shared drive & project board. [P1]	
2		Implemented data_ingestion.py (loading, cleaning, normalization, segmentation); selected & justified ABC News Headlines dataset; wrote prepare_headlines.py. [P1]	
3		Implemented embeddings.py (Sentence-BERT + offline backends); ensured deterministic embeddings; documented preprocessing. [P2]	
4		Built held-out query / sample generation; supported indexing integration testing. [P2/P3]	
5		Data validation for HNSW runs; reproducibility checks on preprocessing seeds. [P3]	
6		Prepared corpora for scalability sizes (1K-100K); verified sizes; logged issues. [P4]	
7		Cross-checked tuning data inputs; assisted results analysis. [P4]	
8		Wrote dataset & preprocessing sections of report; reviewed slides; rehearsal. [P5]	

Ashwin Senthur Pandian — Vector Indexing Lead

Wk	Date(s)	Tasks & contributions	Status / notes
1		Co-wrote proposal (Related Work, Novelty); studied FAISS, IVF and HNSW literature. [P1]	
2		Set up FAISS environment; prototyped the common index interface (BaseIndex). [P3 prep]	
3		Implemented FlatIndex (exact ground-truth baseline) with memory/build-time measurement. [P3]	
4		Implemented IVFIndex (k-means partitioning, nprobe); added nlist auto-scaling for small data. [P3]	
5		Implemented HNSWIndex (M / efConstruction / efSearch); verified all three indexes. [P3]	
6		Tuned default index parameters; supported scalability index builds. [P4]	
7		Supported nprobe / efSearch sweeps; analysed cluster and graph behaviour. [P4]	
8		Wrote indexing methodology & related-work sections; slide content on indexes; rehearsal. [P5]	

Rishi Gaurangkumar Patel — Query & Evaluation Lead

Wk	Date(s)	Tasks & contributions	Status / notes
1		Co-wrote proposal (Objectives, Evaluation Metrics); defined the metric plan. [P1]	
2		Designed the query workload and the evaluation pipeline skeleton. [P2 prep]	
3		Implemented query_engine.py: per-query timing (mean/p50/p95) with warm-up. [P4 core]	
4		Implemented recall@K vs Flat ground truth; results aggregation. [P4 core]	
5		Implemented benchmark.py orchestration (single + scalability); CSV output. [P4]	
6		Ran scalability benchmarks; collected and aggregated all metrics. [P4]	
7		Investigated & resolved recall reproducibility; produced results tables. [P4]	
8		Wrote evaluation & results-analysis sections; slides on metrics; rehearsal. [P5]	

Nikhil Goud Nathi — Experiments, Tuning & Visualization Lead

Wk	Date(s)	Tasks & contributions	Status / notes
1		Co-wrote proposal (System Design, Implementation Plan, Timeline); set up GitHub repository. [P1]	
2		Built run_benchmark.py CLI and the YAML config system. [P3 prep]	
3		Implemented visualization.py (comparative + scalability charts). [P5 tooling]	
4		Integrated the pipeline end-to-end; ran the first full benchmark. [P3/P4]	
5		Built scalability sweep configs; generated scalability charts. [P4]	
6		Implemented tune_parameters.py (nprobe / efSearch sweep) and tuning charts. [P4]	
7		Produced polished presentation figures; reproducibility fixes; tuning analysis. [P4/P5]	
8		Assembled report, README and presentation deck; led rehearsal. [P5]	